

COURSE INFORMATION

Course Code & Title IS216 - Web Application Development II	Effective Start Date 17/08/2026
Offering Unit / Department / School SCIS	Course Career Undergraduate
Course Units CU: 1	Grading Basis GRD - Graded

Course Description

This course is designed to equip students with knowledge and skills to develop well-styled and responsive web applications that provide rich user experiences. Combining with the skills learnt in IS113 course, which focuses on developing database-driven web applications with basic web designs, after this course, the students will be equipped with full stack web development skills, who can build both front-end and back-end software.

In the introductory weeks of the course, the students will revisit HTML and server-side programming (PHP) concepts learnt in IS113. Then, the students will learn the concept of “Styling” the web pages. The students will learn a style sheet language called cascading style sheets (CSS) and learn how to separate the content and presentation of web pages, how to control web page layout, colors and fonts, how to bring multiple styles into a web page, how to control the layout of multiple web pages efficiently, etc. Next, the students will learn the concept of adding responsive behaviors to web pages to enhance the user experience. The students will learn a client-side programming language called JavaScript to make ordinary web elements like input boxes, buttons, forms, tables, menus interactive and animated. Furthermore, they will learn how to connect to API gateways and process data from external sources like RESTful web services so that they can build practical applications.

In the latter weeks of the course, the students will be introduced to programming with frameworks. The students will learn how to use frameworks to build complex web applications in an efficient, scalable manner. More specifically, the students will be introduced to Bootstrap, a popular CSS framework for developing responsive website and introduced to Vue, a progressive JavaScript framework for building rich user interfaces.

Course Learning Outcomes

1. Explain the concept of client-side and server-side programming
2. Understand CSS and be able to create and debug interactive, animated web pages using various styling properties
3. Able to control web page layout, colors and fonts and control the layout of multiple web pages efficiently
4. Understand JavaScript and able to make ordinary web elements like input boxes, buttons, forms, tables, menus interactive
5. Able to manipulate HTML document object model (DOM) and handle events like mouse hovering, mouse click, content changes, etc.
6. Able to connect to API gateways and process JSON data from external sources such as Google map and data.gov.sg
7. Able to use a CSS framework - BootStrap - to build complex web applications in an efficient, scalable manner.
8. Able to use a JavaScript framework - Vue - to build complex web applications in an efficient, scalable manner.
9. Able to solve logic problems that can be encountered while creating web applications, and create suitable solutions using JavaScript and Vue.
10. Able to use several pieces of supporting software together, e.g, WAMP server, Visual studio code, command line interface, Chrome DevTools, etc.

SMU Graduate Learning Outcomes

Disciplinary Knowledge, Critical thinking & problem solving

Competencies

Applications Development, Applications Integration, Problem-solving & analysis, Software development for web applications, Design algorithmic logic for web application interactions

Course Area(s)

AREA - Advanced Business Technology Major, AREA - Business Options, AREA - Econ Major Rel/Econ Options, AREA - Grad Req - Dig Tech/Data Ana (Intake 2024 onwards), AREA - IT Solution Development Electives, AREA - Information Systems Core (Intake 2019 to 2023), AREA - Smart-City Management & Tech Electives, AREA - Social Sciences/PLE Major-related, RQCP - IS113/CS203, TRCK - IS Major: Software Development Track, ENRL - Open to Undergraduates only

Simple Requisites

IS113/CS203 - Pre-req

Type

Prerequisite

Complete ANY of the following Courses:

• IS113 - Web Application Development I

• CS203 - Human-AI Collaborative Software Development

CLASS SECTION INFORMATION

Instructor(s)

SHAR LWIN KHIN - (lkshar@smu.edu.sg)

Course Instructor(s)

Faculty: SHAR Lwin Khin (course manager), SHIM Kyong Jin, MOK Heng Ngee, FWA Hua Leong \n\nInstructor: THIANG Lay Foo, Jason CHUI, WONG Chee Chein, Andrew Tan, Alvin Tay

Instruction Mode

In Person

Assessment Components (New)

Category* Class Participation		Weightage* 15	
Individual or Group?* Individual	With Writing Component?* No	With Oral Component?* Yes	

Category* Final Examination		Weightage* 30		Assessment/Submission Period* Week 15	
Description Final Exam	Individual or Group?* Individual	With Writing Component?* Yes		With Oral Component?* No	

Category* Projects/Assignments		Weightage* 30		Assessment/Submission Period* Week 13	
Description Group Project	Individual or Group?* Group	With Writing Component?* Yes		With Oral Component?* Yes	

Category* Proctored Assessments		Weightage* 25		Assessment/Submission Period* Week 7	
Description Lab Test I	Individual or Group?* Individual	With Writing Component?* Yes		With Oral Component?* No	

Instructional Methods and Expectations

Upon completion of the course, students will be able to:

- understand the concept of styling and adding responsive behaviors to web pages
- build well-styled, responsive web applications using HTML, CSS, and JavaScript
- connect to APIs and process data from external sources using JavaScript
- use existing, available libraries or frameworks to build complex websites in an efficient, scalable manner

5. Competencies

1. Explain the concept of client-side and server-side programming
2. Understand CSS and be able to create and debug interactive, animated web pages using various styling properties
3. Able to control web page layout, colors and fonts and control the layout of multiple web pages efficiently
4. Able to make ordinary web elements like input boxes, buttons, forms, tables, menus interactive
5. Able to manipulate HTML document object model (DOM) and handle events like mouse hovering, mouse click, content changes, etc.
6. Able to connect to API gateways and process JSON data from external sources such as Google maps
7. Able to use a CSS framework – Bootstrap – to build complex web applications in an efficient, scalable manner.
8. Able to use a JavaScript framework – Vue – to build complex web applications in an efficient, scalable manner.
9. Able to solve logic problems that can be encountered while creating web applications, and create suitable solutions using JavaScript and Vue.
10. Able to use several pieces of supporting software together, e.g, Node server, Visual studio code, command line interface, Chrome DevTools, etc.

Weekly Lesson Plan

Week	Topic	Remarks
1	Introduction + HTML/JS Recap + Git	
2	CSS: Properties, Box Model, Positioning	
3	CSS Framework: Bootstrap	Project Briefing; Quiz 1
4	Vue: Building reactive, interactive pages	
5	Vue: API calls, JSON data handling, and Unit testing	Quiz 2
6	Vue: Interaction with backend	
7	Trial Lab Test	Lab Test 1 (25%)
8	Recess	
9	Project Progress Evaluation	5%
10	Vue: Modularization with components and Integration testing	
11	Vue: State Management and E2E testing	Quiz 3
12	Project Work Week (Consultation)	Quiz 4
13	Group Project Presentations	25%
14	Study Week	
15	Final Exam	30%

Recommended Textbooks and Readings

Main Reading:

- Architecting Vue.js 3 enterprise-ready web applications : build and deliver scalable and high-performance, enterprise-ready applications with Vue and JavaScript, Eseme, Solomon, author.; 2023. https://search.library.smu.edu.sg/permalink/65SMU_INST/17n32a3/alma99628487402601

Tools:

- Command Line Interface, Visual Studio Code, Chrome Browser

UNIVERSITY POLICIES

Academic Integrity

All acts of academic dishonesty (including, but not limited to, plagiarism, cheating, fabrication, facilitation of acts of academic dishonesty by others, unauthorized possession of exam questions, or tampering with the academic work of other students) are serious offences.

All work (whether oral or written) submitted for purposes of assessment must be the student's own work. Penalties for violation of the policy range from zero marks for the component assessment to expulsion, depending on the nature of the offence.

When in doubt, students should consult the instructors of the course. Details on the SMU Code of Academic Integrity may be accessed at <https://smu.sharepoint.com/sites/oasis/SitePages/DOS-WKLSWC/UCSC.aspx>.

Responsible Use of AI

The University recognises the importance of equipping students with the skills to navigate an AI-driven world. While AI tools may support learning, they must be used judiciously and responsibly. Students must not present AI-generated output as their own work.

Each course may set specific guidelines on the use of AI in assessments. Students are required to follow these guidelines carefully and should seek clarification from the instructors if uncertain.

Misuse of AI constitutes a breach of academic integrity and will be subject to the same penalties as other acts of academic dishonesty.

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Accessibility

SMU strives to make learning experiences accessible for all. Students who anticipate or experience physical or academic barriers due to disability are encouraged to inform the course instructor as soon as possible. Students may also contact the university's disability services team if they have questions or concerns about academic provisions: accessibility@smu.edu.sg. Please be aware that the accessible tables in our seminar room should remain available for students who require them.

Digital Readiness for Teaching and Learning (DRTL)

As part of emergency preparedness, instructors may conduct lessons online via the Zoom platform during the term, to prepare students for online learning. During an actual emergency, students will be notified to access the Zoom platform for their online lessons. The class schedule will mirror the current face-to-face class timetable unless otherwise stated.